2. Particle Swarm Optimization

Algorithm: PSO_Public_Transport_scheduling.

Begin

Define passenger arrival times and bus scheduling limits

Initialize parameters:

Number of particles N

Inertia weight w

Cognitive coefficient c1

Social coefficient c2

Maximum iterations T

Initialize particle positions X_i and velocities V_i randomly

Evaluate fitness of each particle

Calculate total passenger waiting time.

Fitness = Total waiting time.

Set pBest_i = X_i for each particle

Set gBest = particle with minimum waiting time.

Iteration = 0

While Iteration < T do

For each particle i do

Generate random numbers r1 and r2;

Update velocity:

$V_i = w * V_i + c1 * r1 * (pBest_i - X_i) + c2 * r2 * (gBest - X_i)$

Update position:

$X_i = X_i + V_i$

Ensure bus departure times satisfy constraints

Evaluate fitness of X_i

If Fitness(X_i) < Fitness(pBest_i) then

pBest_i = X_i

If Fitness(X_i) < Fitness(gBest) then

gBest = X_i

Iteration = Iteration + 1

Display gBest bus schedule

Display minimum passenger waiting time.

Fitness Function

To calculate the waiting time for each passenger j, we find the earliest bus departure t_i that occurs after their arrival a_j ($t_i \geq a_j$). The function used to minimize is

$$\min f(X) = \sum_{j=1}^{N} \left(\min_{t_i \geq a_j} (t_i - a_j) \right)$$

$f(X)$ represents total passenger waiting time.

$t_i - a_j$ is the waiting time of passenger j for the next available bus i

Output

Enter number of buses: 3

Enter passenger arrival times in minutes, space-separated:

10 20 35 50

Enter operating start time in minutes: 0

Enter operating end time in minutes: 60

--- Running PSO with Configuration ---

Number of buses: 3

Passenger arrivals: [10, 20, 35, 50]

Operation window: 0 to 60 minutes

Iteration 1/50 | Current Best Waiting Time: 19.65 mins

Iteration 10/50 | Current Best Waiting Time: 10.62 mins

Iteration 20/50 | Current Best Waiting Time: 10.1 mins

Iteration 30/50 | Current Best Waiting Time: 10.05 mins

Iteration 40/50 | Current Best Waiting Time: 10.01 mins

Iteration 50/50 | Current Best Waiting Time: 10.0 mins

--- Optimization Results ---

Optimal Bus Departure Schedule (Minutes from start): [10.0, 20.0, 35.0]

Minimum Total Passenger Waiting Time: 10.0 minutes

```
import random

# --- 1. User Inputs & Configuration with Defaults ---
try:
    # Try to get user input, or fall back to defaults i
    n_buses_str = input("Enter number of buses: ").stri
    n = int(n_buses_str) if n_buses_str else 3

    passengers_str = input("Enter passenger arrival tim
    if passengers_str:
        passengers = list(map(int, passengers_str.split
    else:
        passengers = [10, 15, 25, 40]

    start_str = input("Enter operating start time in mi
    start = int(start_str) if start_str else 0

    end_str = input("Enter operating end time in minute
    end = int(end_str) if end_str else 60
except (ValueError, OSError):
    # Default values for quick execution
    n = 3
    passengers = [10, 15, 25, 40]
    start = 0
    end = 60

print(f"\n--- Running PSO with Configuration ---")
print(f"Number of buses: {n}")
print(f"Passenger arrivals: {passengers}")
print(f"Operation window: {start} to {end} minutes\n")

# --- 2. PSO Parameters ---
N = 20        # Number of particles
T = 50        # Maximum iterations
w = 0.5       # Inertia weight
c1 = 1.5      # Cognitive coefficient
c2 = 1.5      # Social coefficient

# --- 3. Fitness Function ---
# Calculates total passenger waiting time
def fitness(schedule):
    total_waiting_time = 0
    for p in passengers:
        # Find all scheduled buses departing after or a
        valid_buses = [bus for bus in schedule if bus >=
        if valid_buses:
            total_waiting_time += min(valid_buses) - p
        else:
            # Penalty if a passenger misses all buses (
            total_waiting_time += end - p
```

```
            total_waiting_time += end - p
    return total_waiting_time

# --- 4. Initialization ---
# Initialize particle positions (sorted departure sched
X = [sorted(random.uniform(start, end) for _ in range(n

# Initialize particle velocities
V = [[random.uniform(-2, 2) for _ in range(n)] for _ in

# Initialize personal best positions
pBest = [x[:] for x in X]

# Initialize global best position
gBest = min(pBest, key=fitness)[:]

# --- 5. PSO Optimization Loop ---
for iteration in range(T):
    for i in range(N):
        for j in range(n):
            r1 = random.random()
            r2 = random.random()

            # Velocity update equation
            V[i][j] = (
                w * V[i][j]
                + c1 * r1 * (pBest[i][j] - X[i][j])
                + c2 * r2 * (gBest[j] - X[i][j])
            )

            # Position update equation
            X[i][j] += V[i][j]

            # Boundary constraints: Keep bus departures
            X[i][j] = max(start, min(end, X[i][j]))

        # Sort departure times for logical chronological
        X[i].sort()

        # Evaluate fitness and update personal best
        if fitness(X[i]) < fitness(pBest[i]):
            pBest[i] = X[i][:]

        # Evaluate fitness and update global best
        if fitness(X[i]) < fitness(gBest):
            gBest = X[i][:]

    # Optional: print convergence progress every 10 ite
    if (iteration + 1) % 10 == 0 or iteration == 0:
        print(f"Iteration {iteration + 1}/{T} | Current

# --- 6. Results ---
```

```
print("\n--- Optimization Results ---")
print("Optimal Bus Departure Schedule (Minutes from sta
print("Minimum Total Passenger Waiting Time:", round(fi
```

```
Enter number of buses: 3
Enter passenger arrival times in minutes, space-separated: 10 20 35 50
Enter operating start time in minutes: 0
Enter operating end time in minutes: 60

--- Running PSO with Configuration ---
Number of buses: 3
Passenger arrivals: [10, 20, 35, 50]
Operation window: 0 to 60 minutes

Iteration 1/50 | Current Best Waiting Time: 27.28 mins
Iteration 10/50 | Current Best Waiting Time: 11.02 mins
Iteration 20/50 | Current Best Waiting Time: 10.2 mins
Iteration 30/50 | Current Best Waiting Time: 10.03 mins
Iteration 40/50 | Current Best Waiting Time: 10.01 mins
Iteration 50/50 | Current Best Waiting Time: 10.0 mins

--- Optimization Results ---
Optimal Bus Departure Schedule (Minutes from start): [20.0, 35.0, 50.0]
Minimum Total Passenger Waiting Time: 10.0 minutes
```